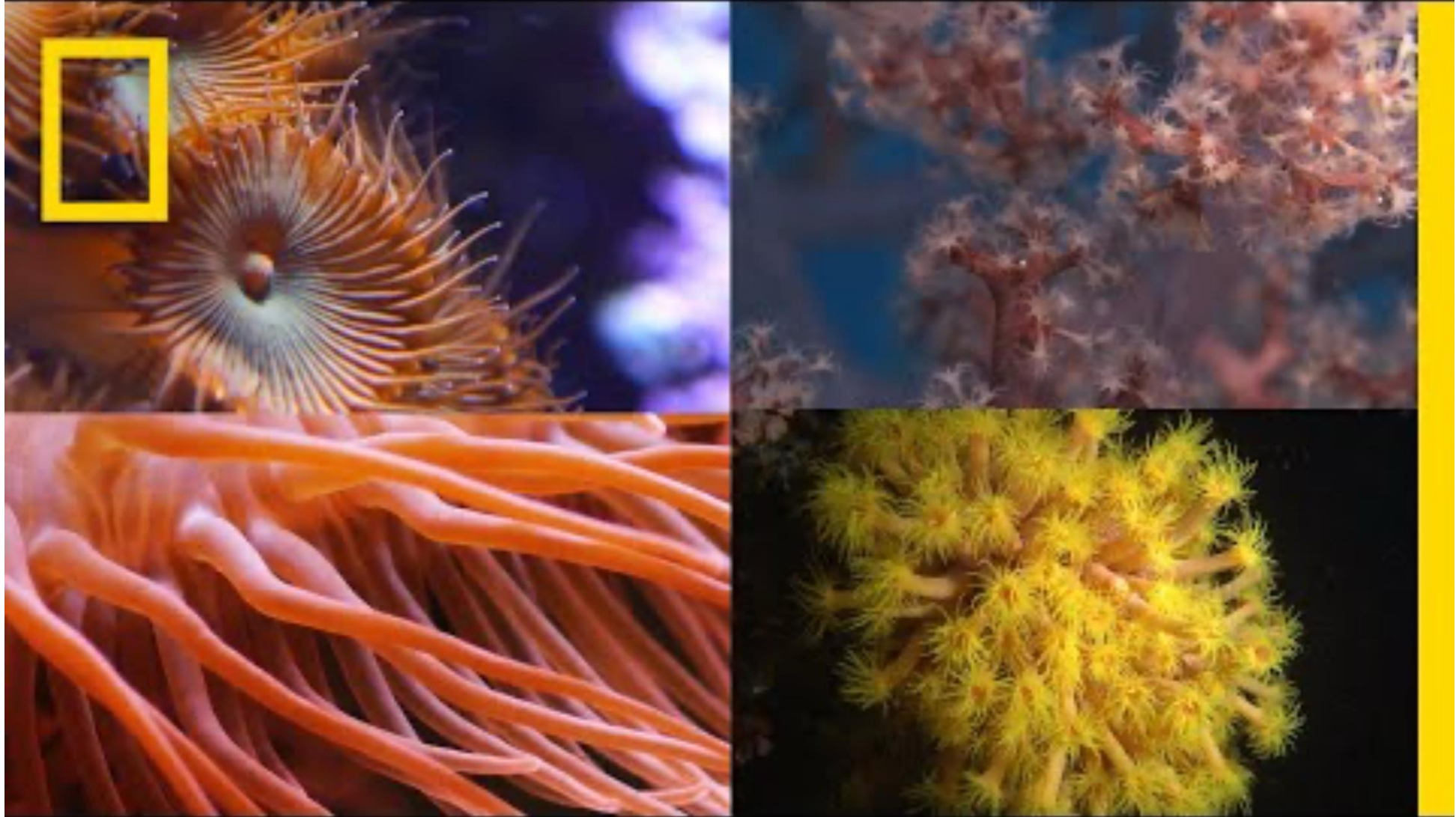


CORALS DAY 2!



Yesterday we started to understand what conditions corals need to thrive. Today we are going to better understand why the Great Barrier Reef is 1/2 the size than it was about 20 years ago....

2016



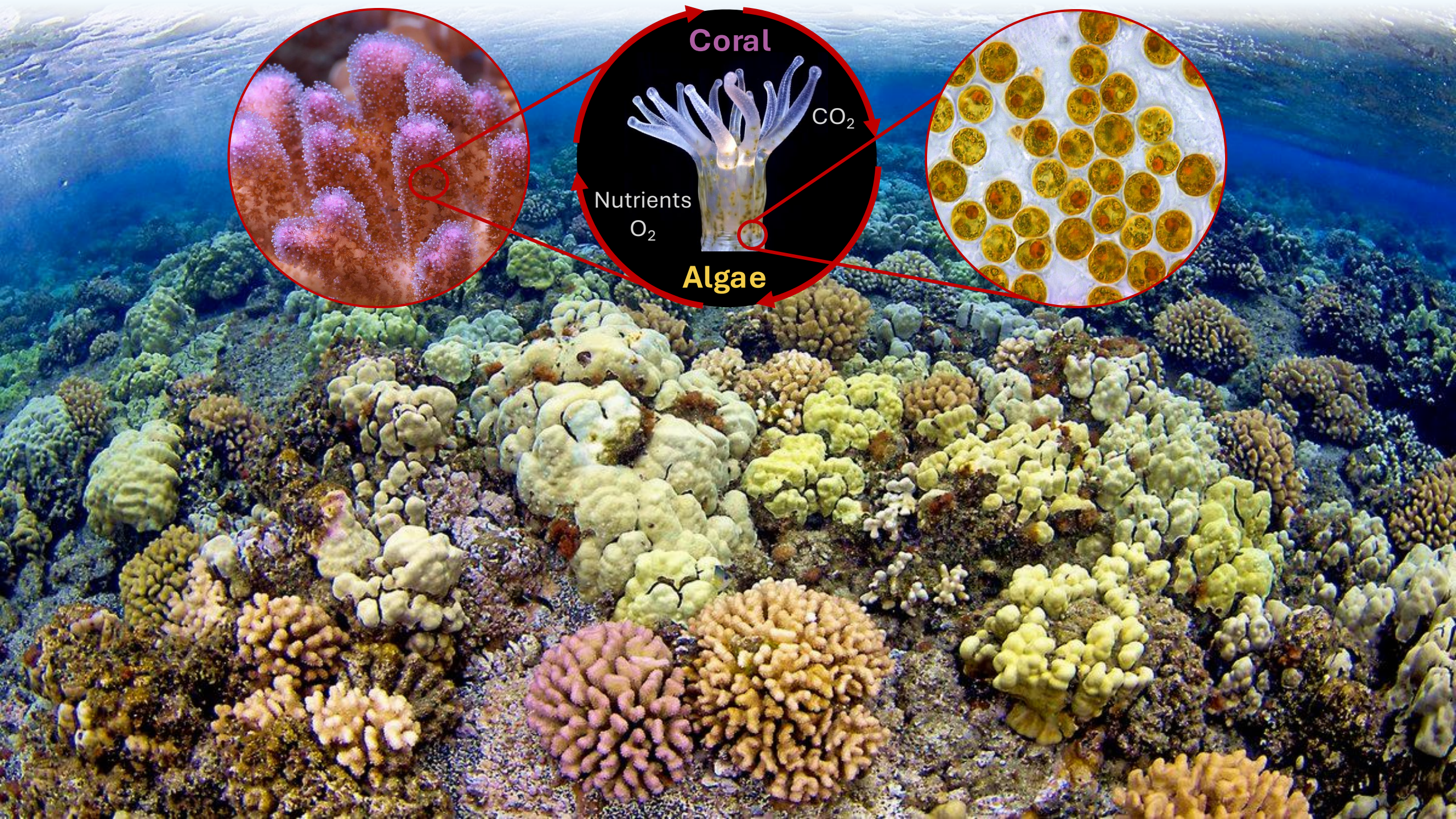
2017



The background features stylized blue waves in the top right and bottom left corners. Several small, light blue fish are swimming in the upper right and lower left areas.

02

Corals are plants



Coral

CO₂

Nutrients

O₂

Algae

Corals still
have algae
in their
tissue



Corals are white
because they
have lost their
algae



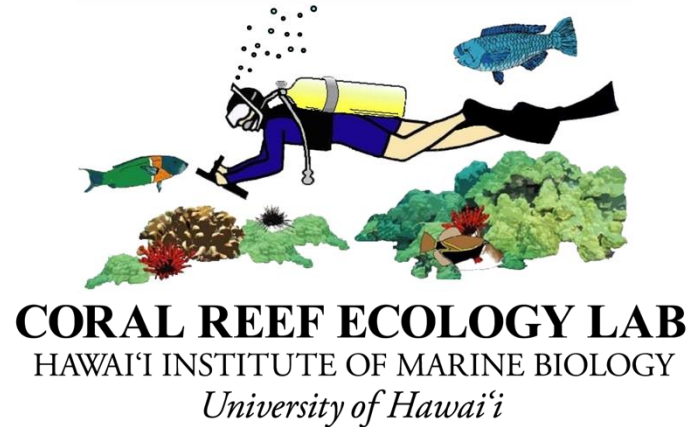
Healthy VS Bleached

Catlin seaview survey



The Hawaiian Ko'a Card

a coral health assessment tool



How to use the Ko'a Card: Let's Survey!

Survey



Coral Color



Report

Hawaiian Ko'a Card: Coral Health Assessment Tool

DATA ENTRY FORM

*Coral Color:

*Coral Shape:

Coral Species:

show scientific names

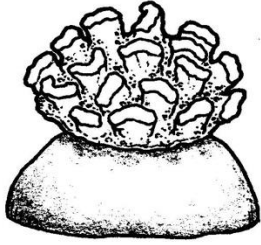
Coral Bleaching:

Depth:

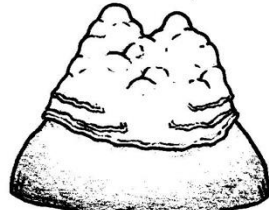
Water Temperature:

Coral Shape

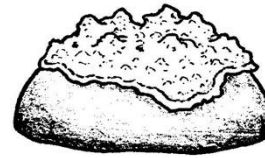
BRANCH



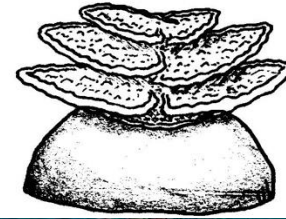
MOUND



CRUST

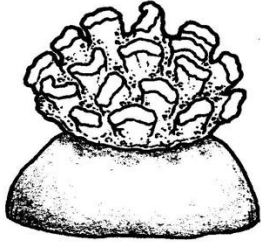


PLATE

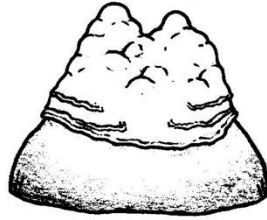


What shape is this color?

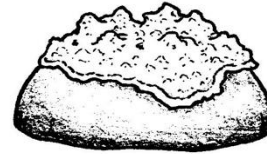
BRANCH



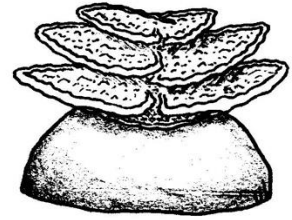
MOUND



CRUST

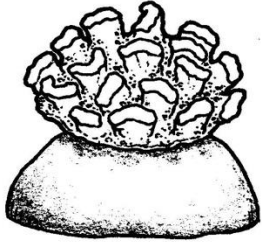


PLATE

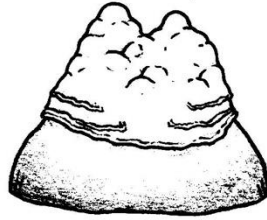


What shape is this color?

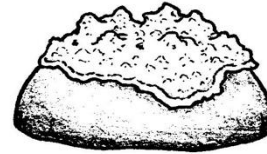
BRANCH



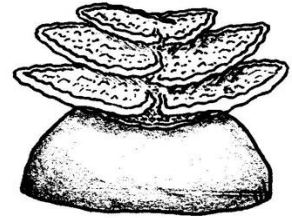
MOUND



CRUST



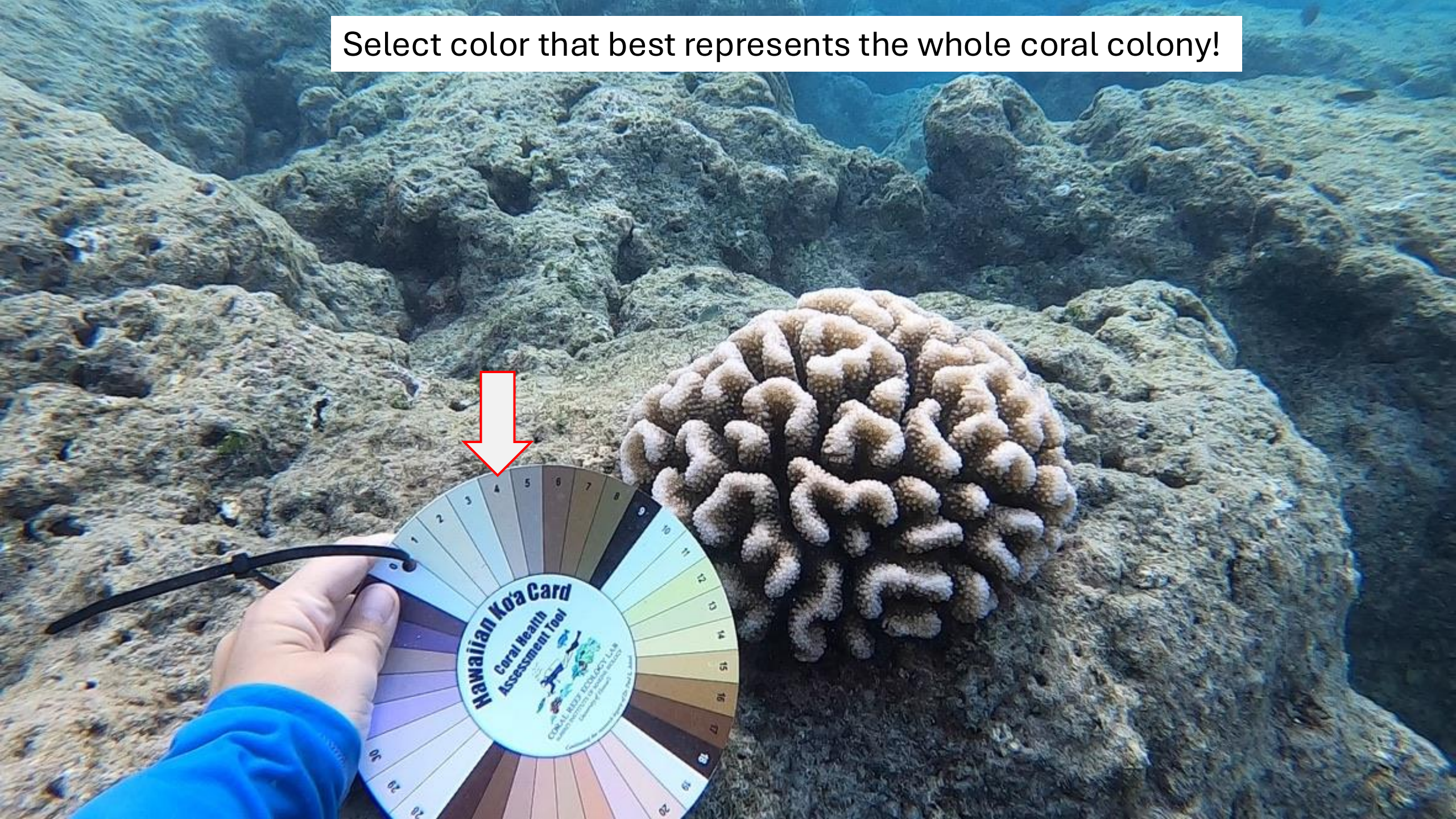
PLATE

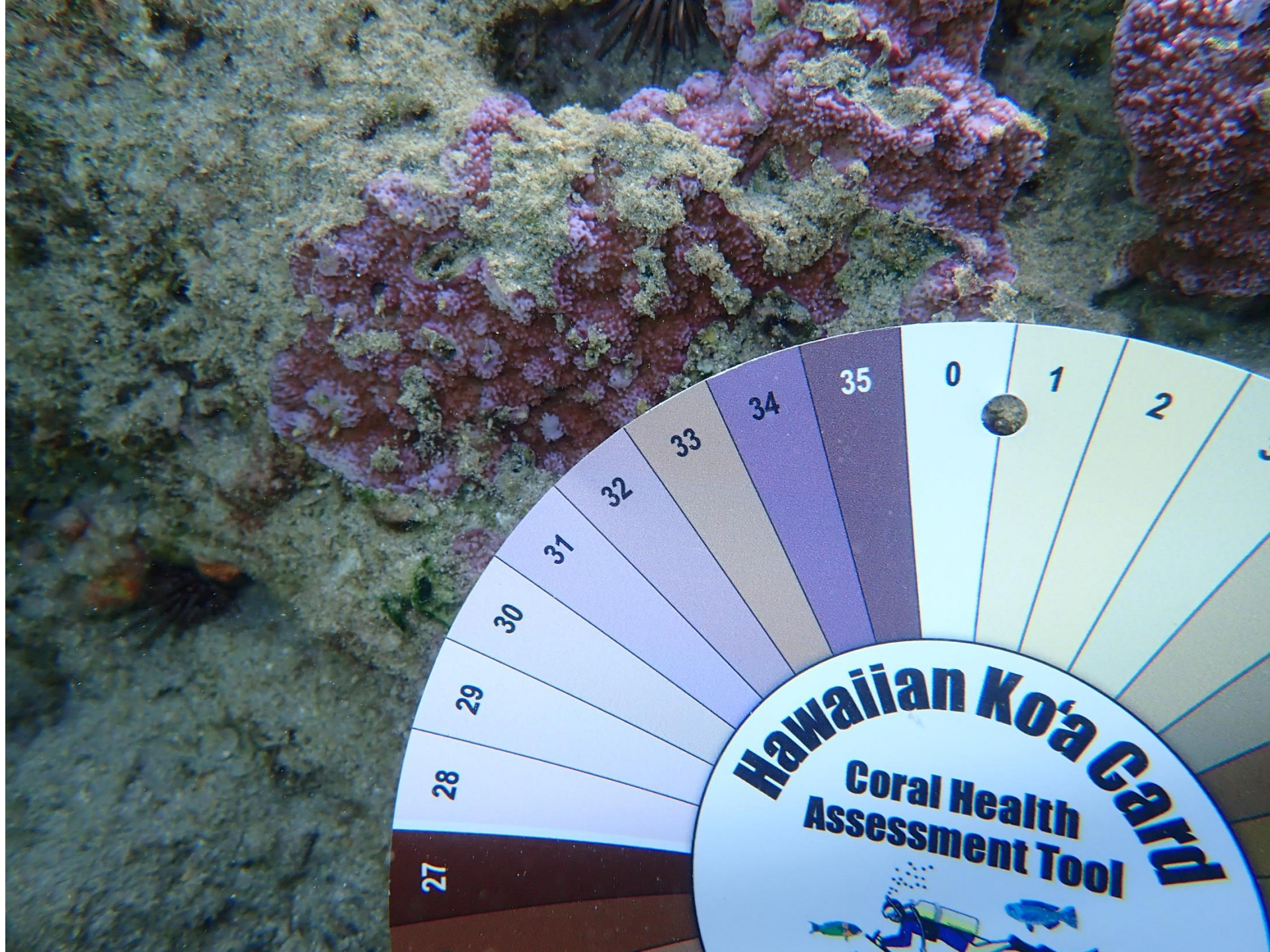
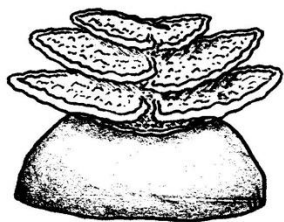
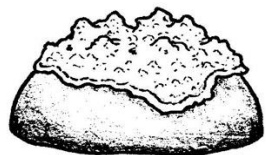
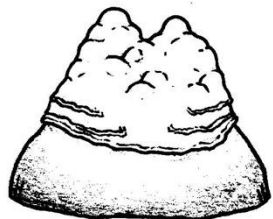
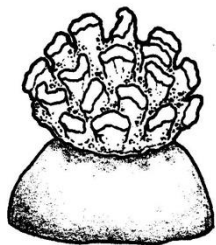


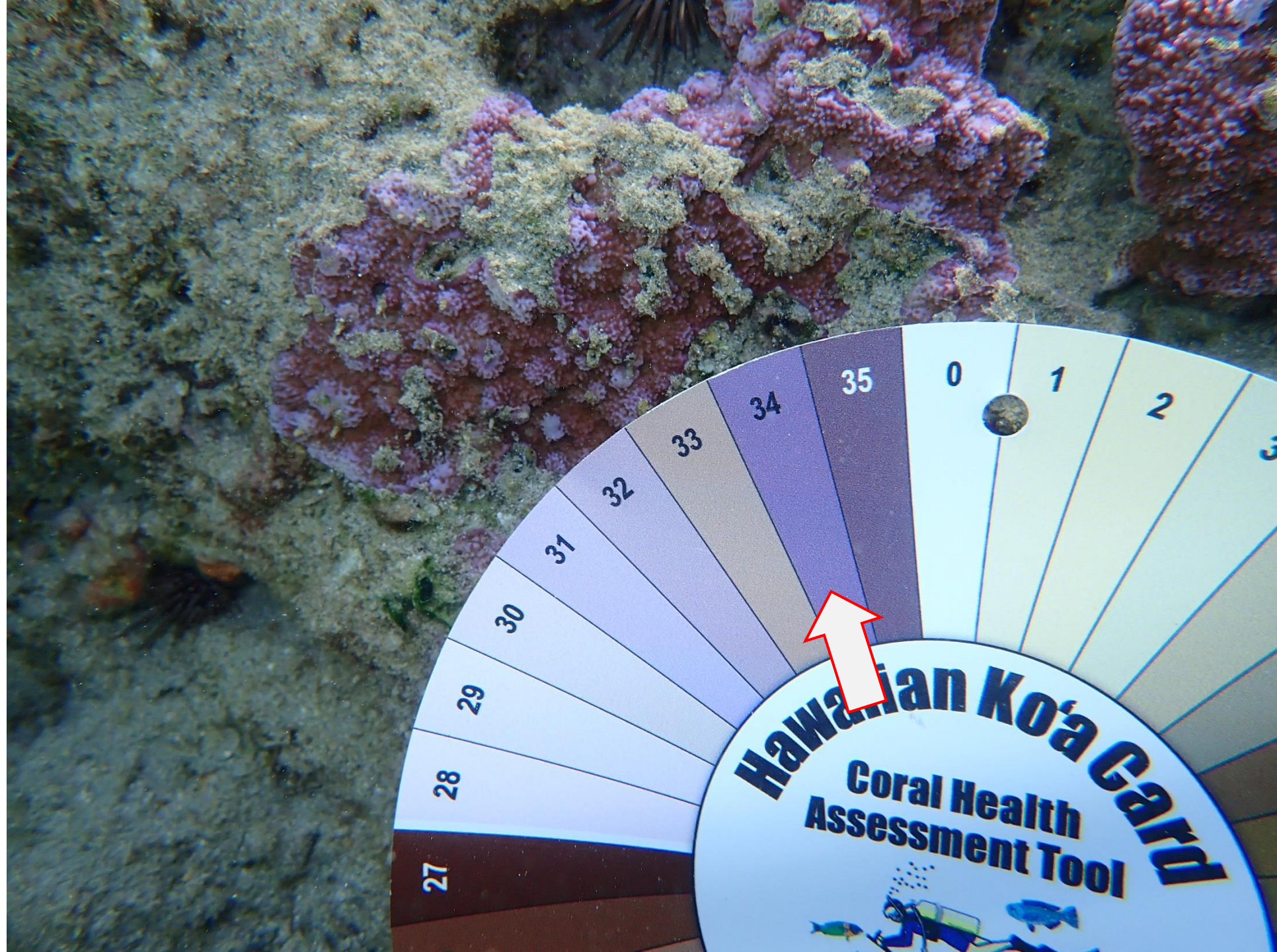
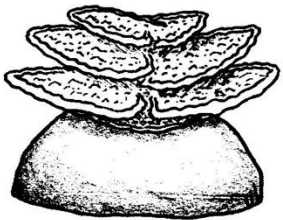
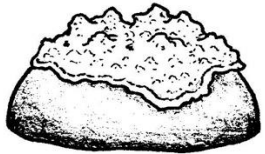
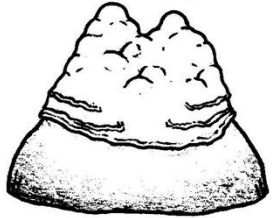
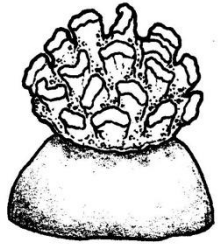
Select color that best represents the whole coral colony!



Select color that best represents the whole coral colony!









03

Corals are rocks

What is a reef?

- A reef = hard bottom structure composed by organisms and used by other organisms
 - Reefs themselves are not alive
- A **Coral Reef** is a reef build from the carbonate skeletons of corals



Coral Growth (Calcification)

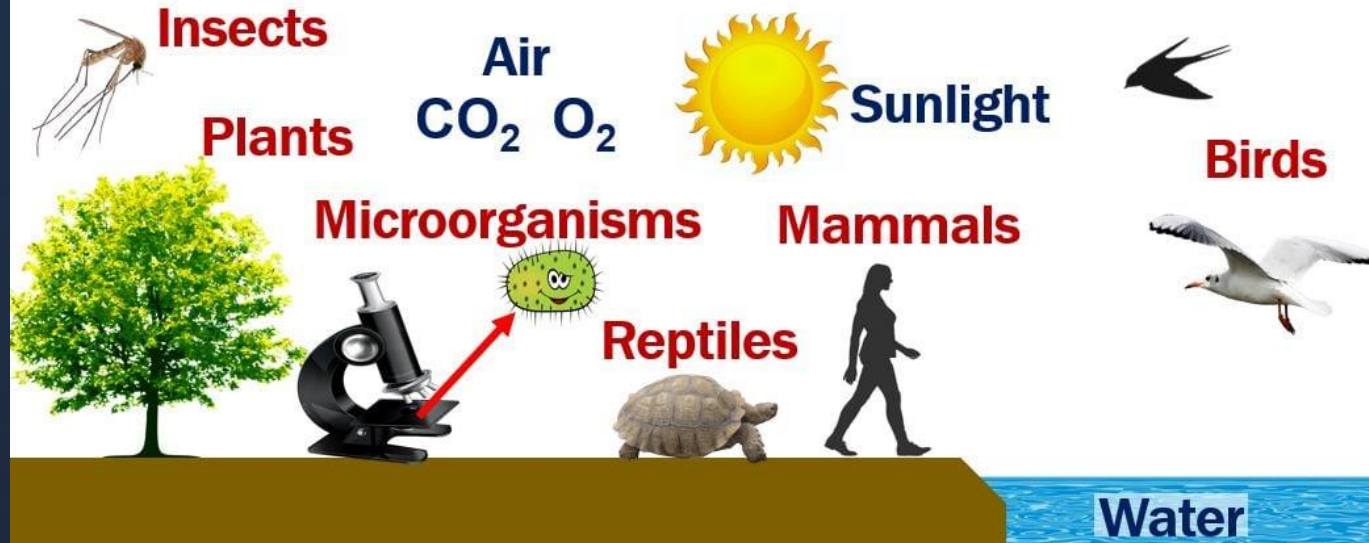
- Calcification = Process by which corals produce calcium carbonate (CaCO_3) skeleton
- 30% of corals' energy budget is devoted to calcification
- Calcification is accelerated by light = photosynthesis-driven process
- Increases surface area & structural support for the organism
- A way of maintaining a close proximity to high light levels
- Means for keeping up with sea level rise



Ecology

The study of the relationships between organisms and their environment, and the balances between these relationships

Blue Words = Abiotic Factors Red Words = Biotic Factors



Just like you have an environment, corals also live in an environment which have certain living and nonliving factors that make up their ecosystem

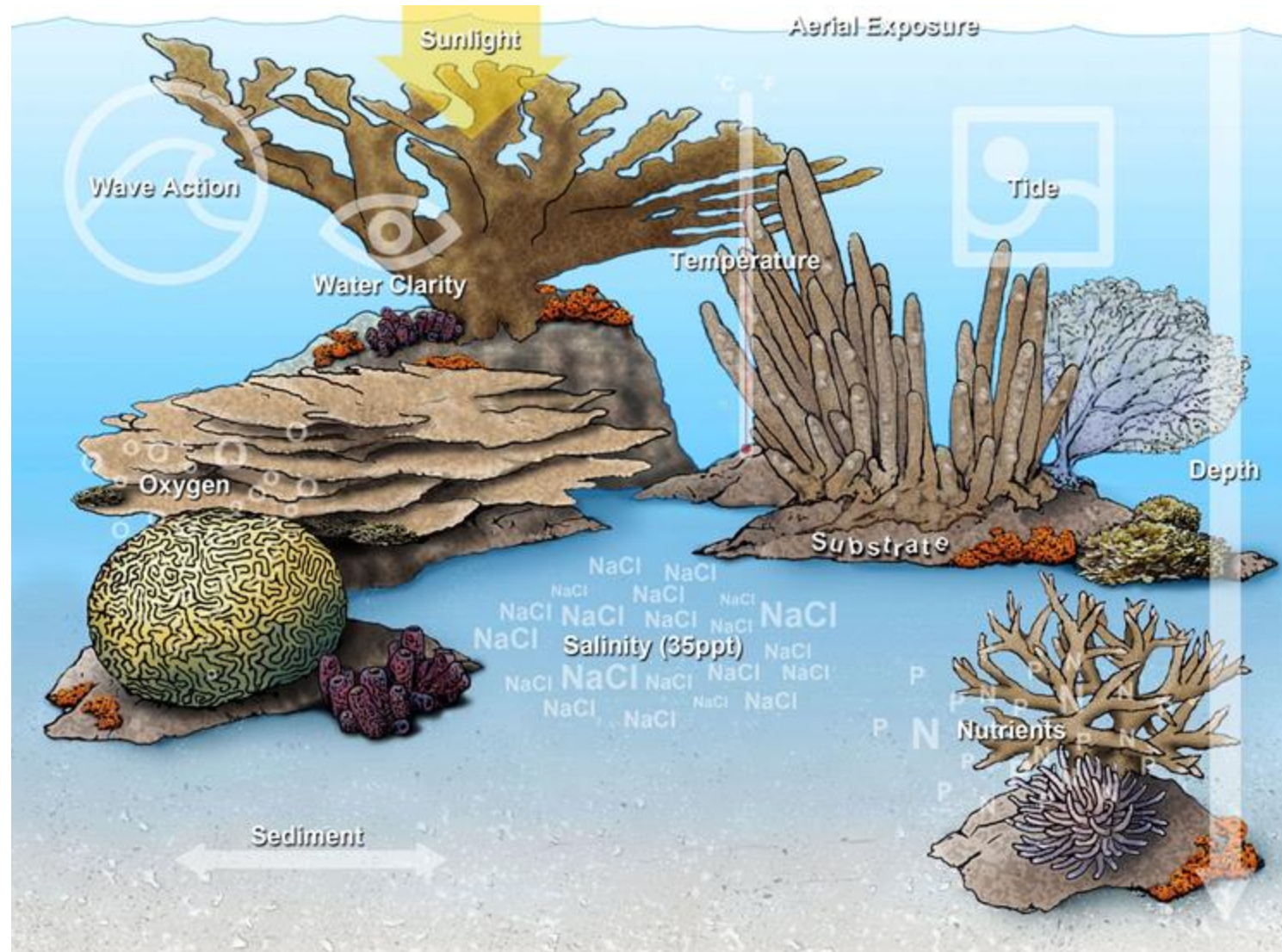
Ecosystem

An ecosystem is a geographic area where plants, animals, and other organisms, as well as weather and landscapes, work together to form a bubble of life.

Biotic Factors



Abiotic Factors



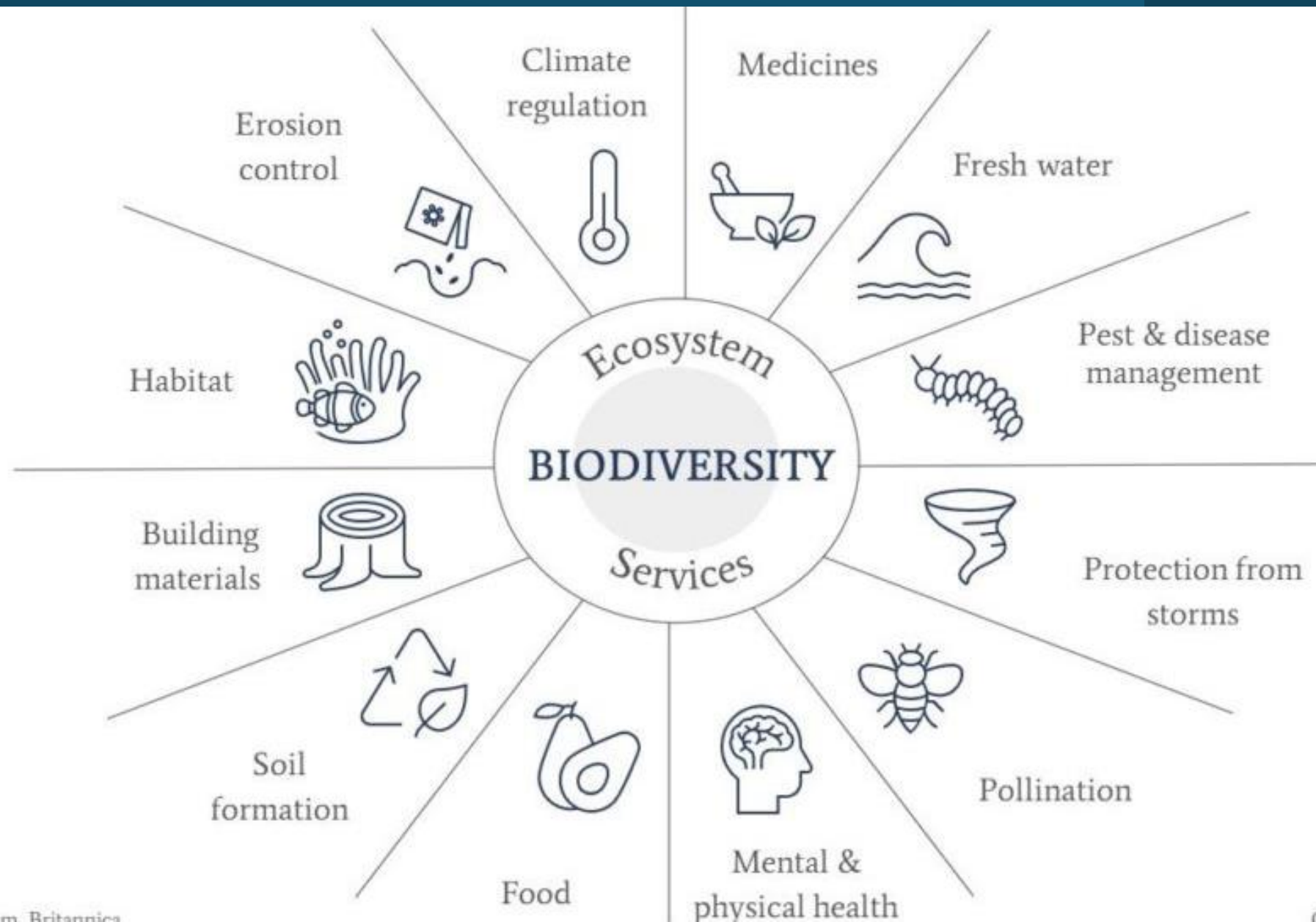
WHAT IS BIODIVERSITY?



Biodiversity refers to the variety of living organisms on Earth and the interactions between them. This includes the diversity of species, ecosystems, and genetic variations within a species.

Biodiversity is important because it provides numerous benefits to humans, such as food, medicine, and ecosystem services like pollination and water filtration. It also plays a crucial role in maintaining the balance and resilience of ecosystems, which in turn supports the sustainability of human life.

However, **biodiversity** is currently threatened by various human activities, including habitat destruction, pollution, and climate change. Conservation efforts are needed to protect and restore



Coral Reefs = less than 1% of the ocean & home to over 25% of marine life

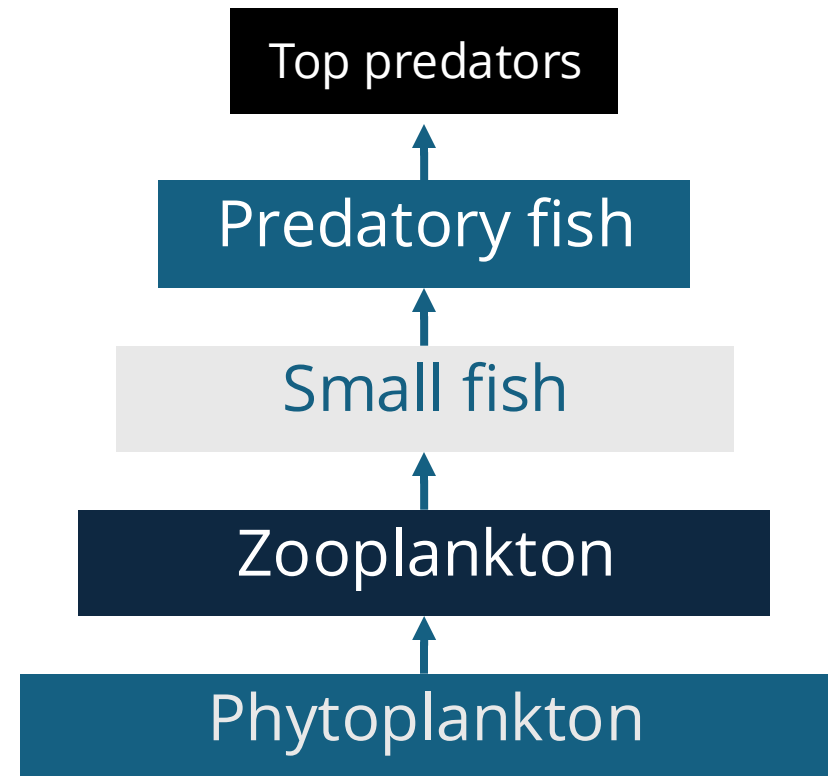
Coral reefs are tropical marine ecosystems that support a high diversity of plants and animals. Because of their exceptionally high species diversity, scientists often refer to coral reef as the tropical rainforests of the oceans.



The ocean food web = **A delicate balance**

The Ocean Food Web = **complex interactions among marine species**

- Essential for **sustaining and diversity of our oceans**
- Showing the transfer of energy and nutrients through various trophic levels
- Delicate balance of predator and prey, each species playing a vital role in the functioning of marine ecosystems
- Reminding us of the interconnectedness of all living organisms in the ocean



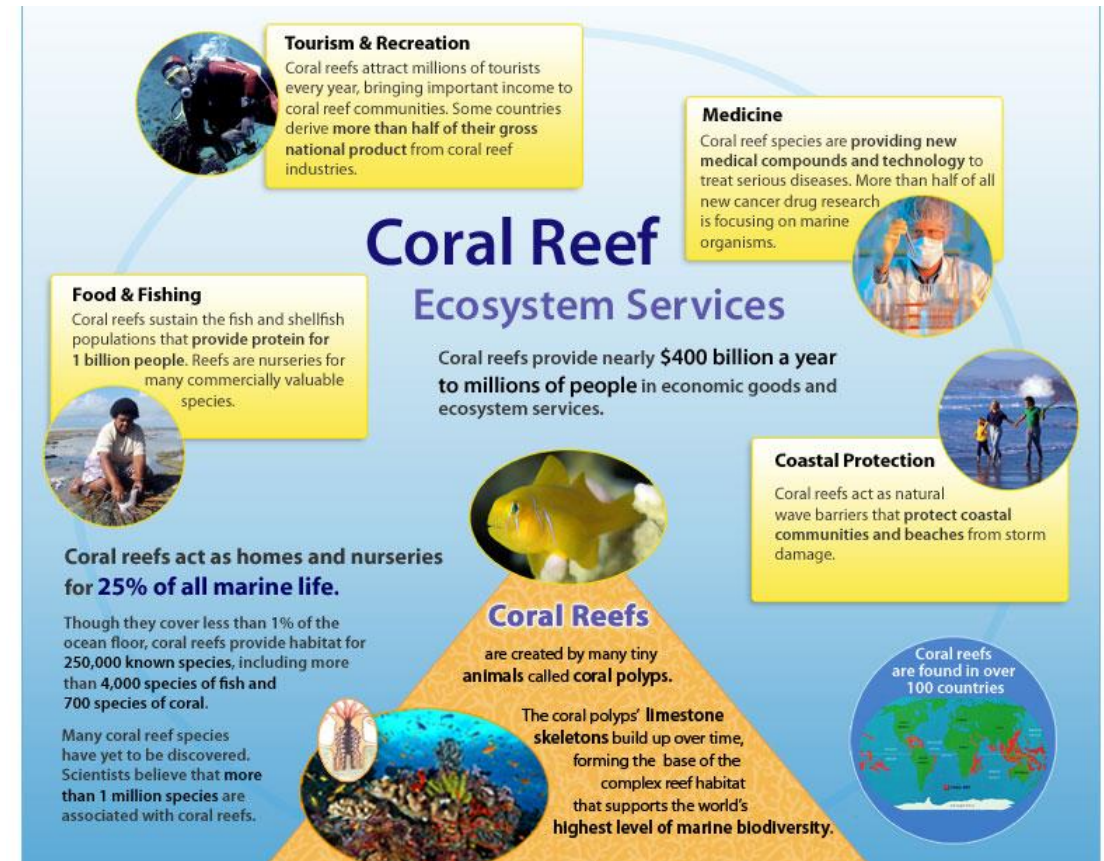


Why do we care about coral reefs?



Definition: Ecological Services

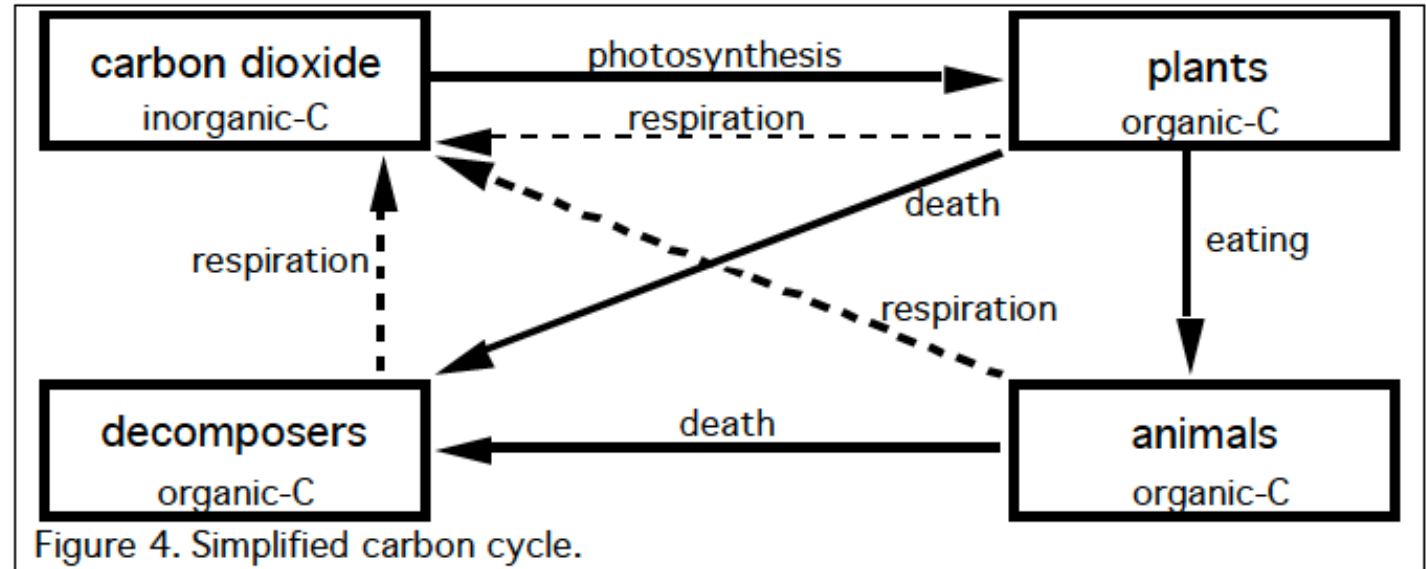
- ▶ The natural world provides us with the essential services we require for life.
- ▶ These services are called ecological services - the interactions among organisms and their natural environments, including the cycling of water and basic nutrients that humans can use and capitalize on.



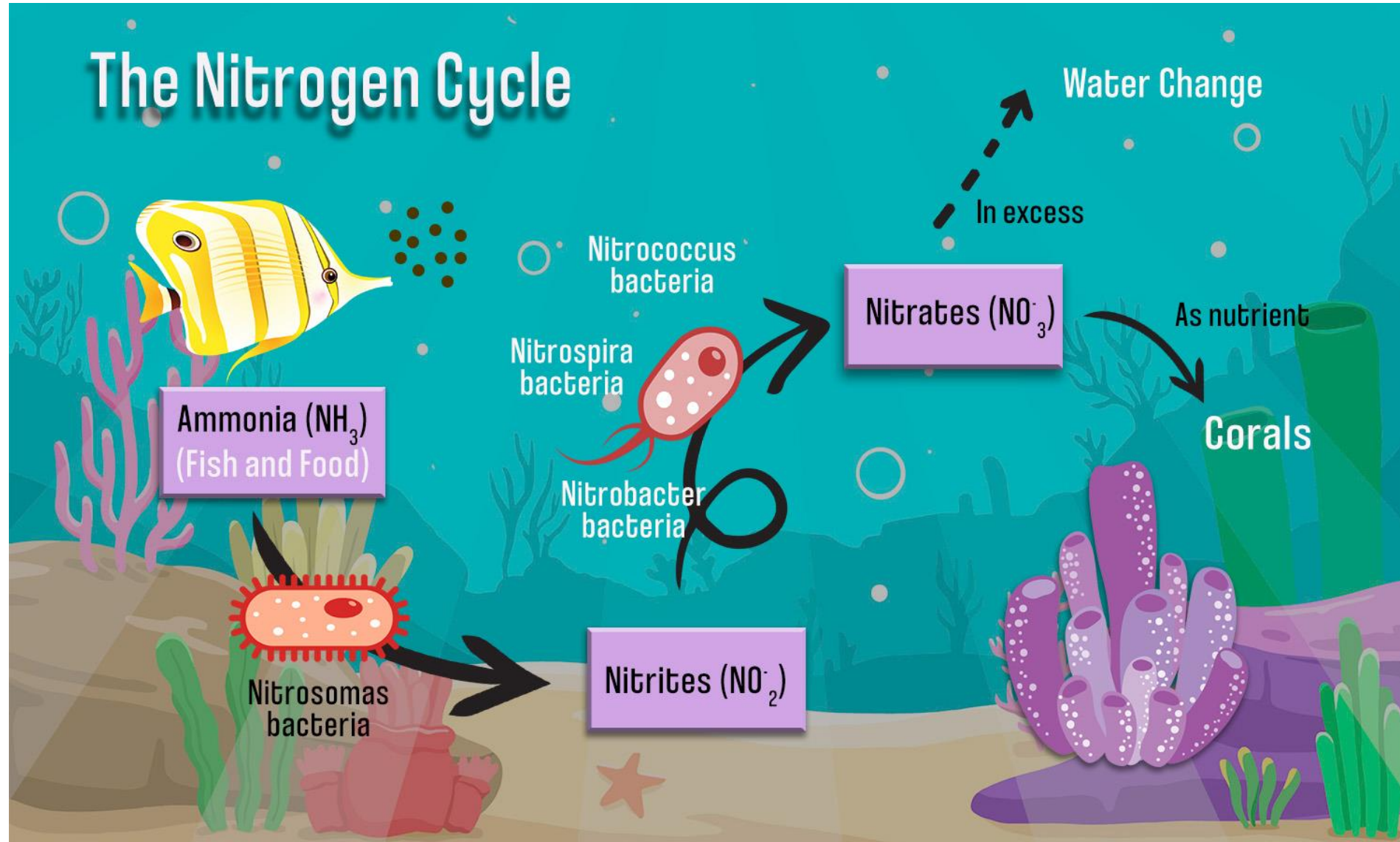
Biogeochemical Cycles on Coral Reefs

Carbon & Nitrogen Cycle!

While energy fluxes through ecosystems, materials are **recycled and regenerated**



Biogeochemical Cycles on Coral Reefs



Carbon cycle

- Corals can **capture, release and accumulate** carbon in different forms
- Carbon sink = absorbing CO₂
- To determine corals' **carbon balance**, one way is to consider the **difference between the quantities of absorbed and released carbon during corals' metabolism**
- **Ocean acidification linked to the human-driven increase of atmospheric CO₂ jeopardizes the equilibrium between these processes**

Carbon cycle disrupted – OCEAN ACIDIFICATION



Carbon Cycle disrupted by Climate Change

Warming

Coral Bleaching

Acidification

Weaker skeletons

More intense storms

Coral's breaking

UV radiation

Ozone depletion and
increased light

Ocean Circulation Patterns

Changing nutrient inputs onto reefs

Less Fish

Fish eat the algae on the
reef so corals can grow

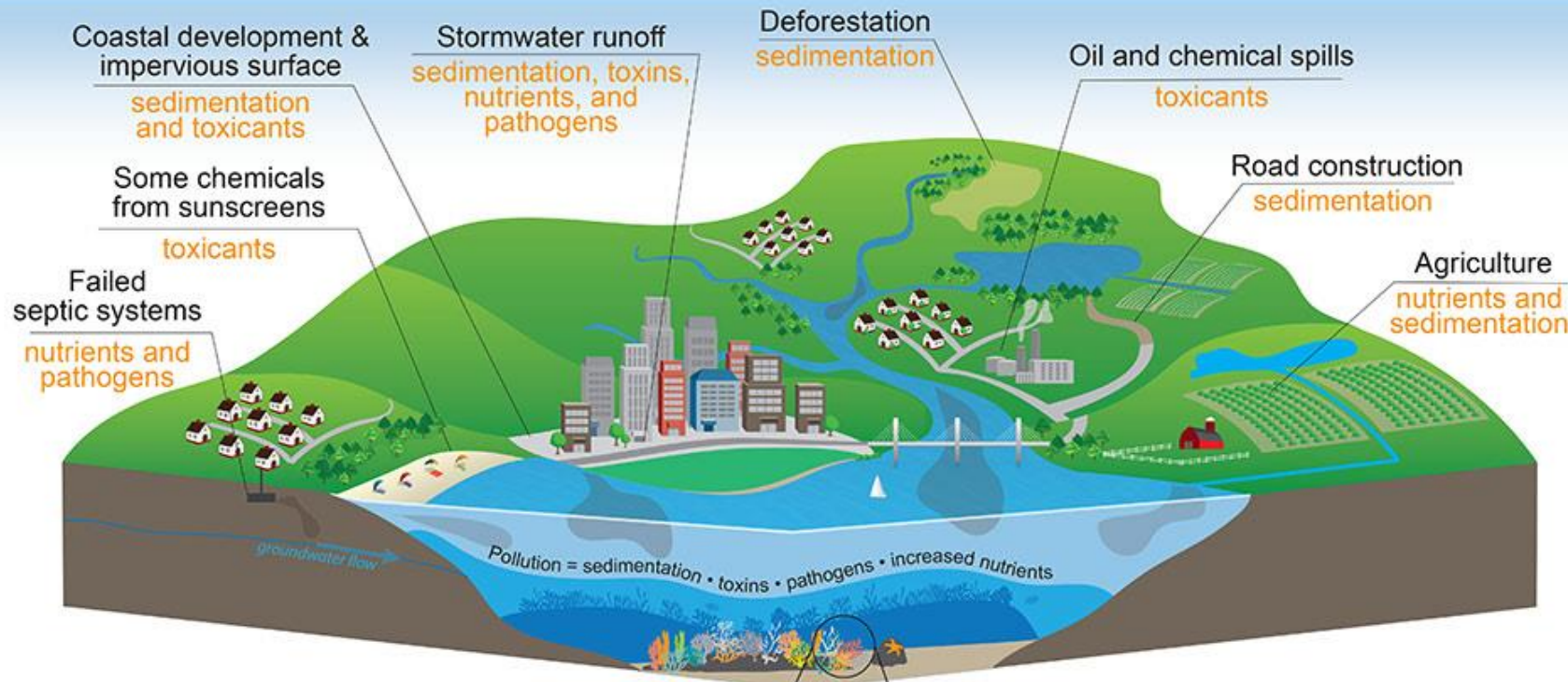


WHY ARE CORAL REEFS DYING?

EPISODE
3 OF 9



THREATS TO CORAL REEFS LAND-BASED SOURCES OF POLLUTION



POLLUTION

sedimentation
toxicants
pathogens
increased nutrients

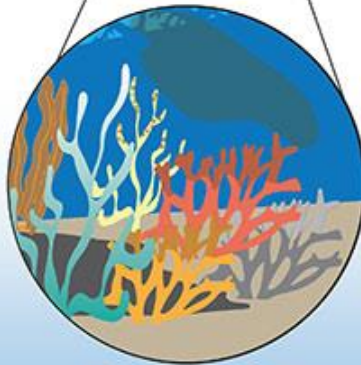


causes disease and mortality

disrupts ecological functions

changes dynamics and feeding behaviors

prevents coral growth and reproduction



As human population & development expands in coastal areas, the landscape is altered, increasing land-based sources of pollution & **THREATENING CORAL REEF HEALTH.**

HOW YOU CAN HELP

Apply fertilizers and pesticides sparingly.



Pick up after your pets.



Wash your car on your lawn.



Dispose of lawn clippings in a compost pile.



Harvest rooftop rain water through rain barrels or rain gardens.



DO NOT dump paint, oil, antifreeze, debris, or other household chemicals into street gutters or storm drains.



Clean up spilled brake fluid, oil, grease, and antifreeze.



Maintain proper septic system function with inspections and pumpouts every 3-5 years.

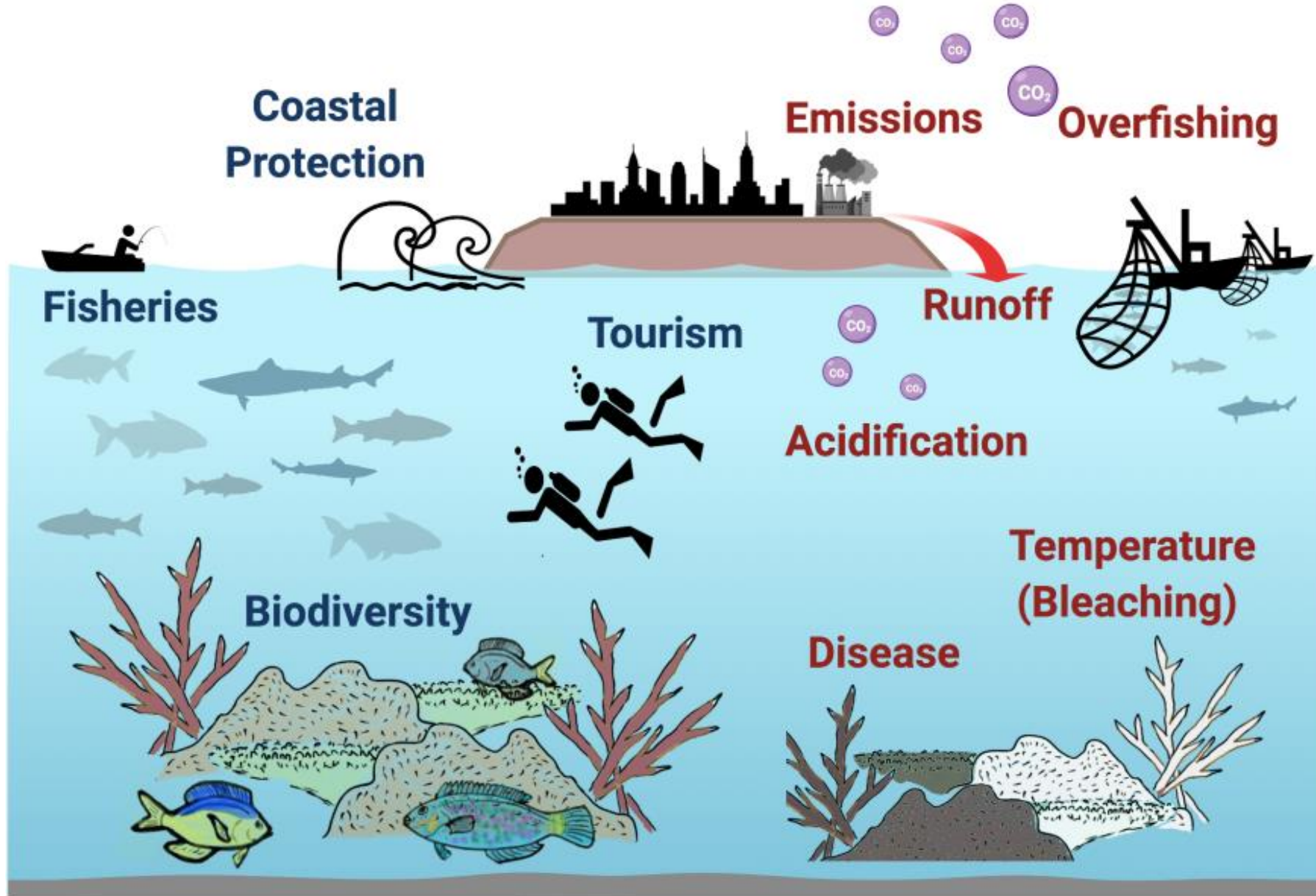


Seek shade between 10 a.m. and 2 p.m., use Ultraviolet Protection Factor (UPF) sunwear, and choose sunscreens with chemicals that don't harm marine life.



For more information, visit oceanservice.noaa.gov/sunscreen.

Ecosystem Services & Threats





The future of our oceans

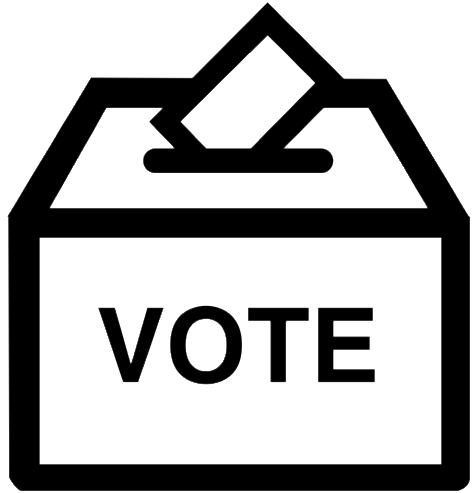
- **Climate change:** Our oceans face the ominous threat of climate change, resulting in rising sea levels, ocean acidification, and increasingly frequent extreme weather events. These changes have far-reaching consequences for marine life and coastal communities alike
- **Overfishing:** Unsustainable fishing practices have depleted fish stocks and strained marine ecosystems. The challenge is to transition toward responsible, sustainable fisheries to prevent the collapse of crucial fishery resources
- **Pollution:** From plastic waste to oil spills, our oceans continue to suffer from pollution. We must take decisive action to mitigate the harmful effects of pollutants on marine life and ecosystems
- **Habitat destruction:** Coastal development and habitat destruction disrupt delicate marine ecosystems, impacting biodiversity and vital breeding grounds



Take Action!

Love Your Oceans By Taking Care of Them

Vote for change!



Support Green Energy



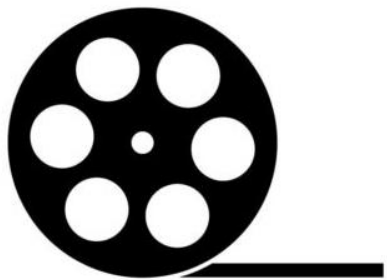
Reduce Carbon



Educate Yourself



Share Chasing Coral



Reduce Plastic Use



Support Conservation



Share Knowledge

